



Metal that doesn't conduct heat

Researchers have identified a metal, Vanadium Dioxide, that doesn't conduct heat, but conducts electricity. <https://digit.in/dec19-65>



Majungasaurus' teeth

The approximately 70 million year ago Majungasaurus replaced all its teeth every two months. <https://digit.in/dec19-66>

gist recorded a single crack appearing on the surface in Massachusetts, United States. The local residents reportedly heard a loud booming noise in the early hours of the morning. Hitchcock surmised that a nearby stream flooding had saturated the ground with water, and then the freezing temperatures had caused the earth to crack. This is essentially the process by which frostquakes are believed to occur today.

The energy produced by these quakes are not enough to register on seismographs. At times the electric charges built up within the rocks can be released following a frostquake, leading to flashes of lights close to the surface. The frostquakes are accompanied by very low frequency vibrations, and do not take place along the fault lines, which see the most earthquakes.

The process by which a frostquake occurs is illustrated in the diagram shown below. In panel 1, the water precipitates into the ground, or can accumulate after flooding. With a drop in temperature, the water begins to turn into ice, as shown in panel 2. As the temperatures continue to drop, the ice begins to expand, exerting pressure on the surrounding soil and rock, illustrated by the red arrows in panel 3. Finally, when the ground can no longer tolerate the mounting pressure, it is released by the formation of cracks, as shown in panel 4. These cracks can extend to the surface, and result in a loud booming noise. If the snow is tightly packed above the ground, then a frostquake does not take place. However, if there

is little to no snow at the ground level, then a frostquake can occur.

We now know that the conditions required to induce a frostquake requires first the liquid water to permeate into the ground, and to subsequently freeze up. During the winter months, if the temperature rises above zero, then this can cause the accumulated snow to melt. The melting snow then quickly enters the ground. As the temperatures fall again, a frostquake can occur. A study conducted by scientists in Canada shows that over the last sixty years, the number of days where the temperature rises above zero has been increasing. The recorded rate of increase is roughly about one day every five years. At the same time, the number of no snow days are also increasing. This means that the ground is not covered with a layer of snow, which prevents the frostquakes from occurring. While the frequency of recorded frostquakes in Canada and the United States have been increasing in recent years, the relationship with global warming is tenuous, and as such is a contentious issue.

ICEQUAKE

This geological activity is even more obscure than frostquakes, and has only been observed in Greenland, using very sensitive instruments. Like the frostquakes, the icequakes also are accompanied by low frequency vibrations, and are not associated with fault lines. Unlike the frostquakes though, these Greenland icequakes take place in the warmest months of the year, in July,

August and September. These are the months where the most amount of glacial ice melts. The earthquakes caused by tectonic activity do not exhibit a seasonality, like the icequakes do.

The melting water is believed to seep into the ground, and accumulate in pools below the glacier. If the pool grows large enough, it acts as lubrication for the glacier, and can cause the entire glacier to slip,

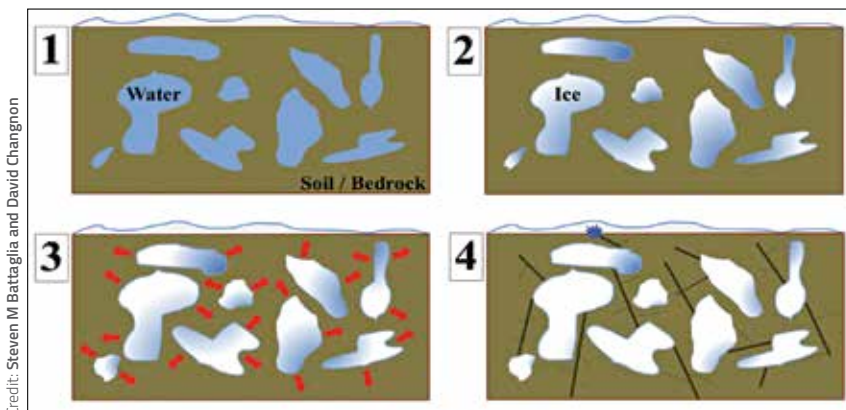


Credit: Aigkaly

The Russel Glacier in Greenland

similar to how a person would slip on a tile with a thin film of water on top. These icequakes are by no means as localised as the frostquakes.

A geologist from Harvard called Göran Ekström has recorded one instance where 100 cubic kilometres of glacier slipped by 12 meters within a minute. The low frequency waves are caused by the ice grinding against the rocks below. If there is a gradual slope, the glacier can continue to slip for longer durations. Eventually the friction builds up, causing the glacier to stop. Such large scale slipping of glaciers can actually induce an earthquake. If a portion of a glacier as large as 60 square kilometers and 400 meters high slips by 10 meters, it can cause a magnitude 5 earthquake. A team led by Ekström have studied instances of icequakes since 1993, and have noticed that the frequency of icequakes have also been increasing over the years. The number of yearly icequakes recorded has almost doubled in number from the early 1990s to the mid 2000s. The increase in frequency of icequakes closely tracks the increase in temperature in Greenland. <https://digit.in/dec19-67>



The process behind a frostquake

Credit: Steven M Battaglia and David Changnon